Yen-Hsiang (Robert) Huang

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Educations

Ph.D. in Electrical and Computer Engineering | GPA: 4.0/4.0 Sept. 2022 - Present
Georgia Institute of Technology (Gatech), Atlanta, GA, USA
B.S. in Electrical Engineering | GPA: 4.06/4.30 Sept. 2017 - July 2022
National Taiwan University (NTU), Taipei, Taiwan

Research Experience

Graduate Research Assistant
GTCAD Lab., Georgia Institute of Technology, Advisor: Prof. Sung Kyu Lim Sept. 2022 - Present

3D IC Optimization Fall 2024 - Present
- Timing optimization through reduced critical-path snaking and inter-die cell relocation (Snake-3D).
- Developed an RC-calibrated differentiable clock-tree optimization framework for 3D ICs, incorporating interconnect RC effects into timing-aware clock-tree optimization (CLK-3D).

Semiconductor Research Corporation (SRC) – JUMP 2.0 CHIMES Fall 2023 - Spring 2024
- Bridged the PPA gap between legacy-node 3D ICs and advanced-node 2D ICs.
- Reduced WNS of legacy-node 3D ICs by 50–93% compared to SOTA, achieving 87.9–97.7% of advanced-node 2D performance.

Die Bonding Bumps and Pads Legalization for 3D ICs Sept. 2022 - May 2023
- Developed a 3D-via legalization stage with a refinement step while minimizing runtime impact; see the related TCAD journal article and ISPD conference paper.
- Eliminated all 3D-via overlaps when via utilization is less than 40%.

Individual Study
EDA Lab., National Taiwan University, Advisor: Prof. Yao-Wen Chang Apr. 2020 - Sept. 2021

Independent Study on Physical Design
- Participated in physical-design-related global contests three times across different teams and placed in the top five twice (ISPD contest 2021 and ICCAD contest 2021).

Individual Study
ALCom Lab., National Taiwan University, Advisor: Prof. Jie-Hong Jiang Sept. 2020 - June 2021

Independent Study on Reversible Circuit Synthesis
- Implemented a reversible circuit synthesizer for controlled-NOT (CNOT) gates.

Work Experience

R&D Engineer Intern **Sunnyvale, CA**
Synopsys Inc. May 2025 - Dec. 2025
- Investigated machine-learning-based signal-integrity (SI) analysis for 2.5D ICs, evaluating multiple models to predict final-stage SI performance from early-stage design data.
- Independently implemented two provided prototypes in C++ and merged the resulting code into the main branch.

Software Engineer R&D Intern **Hsinchu, Taiwan**
Synopsys Inc. - Concurrent Clock-and-Data (CCD) Team July 2021 - Aug. 2021
- Enhanced the flip-flop merging algorithm in Synopsys IC Compiler II (ICC II) by considering the flip-flop's timing (slack)
- Improved the timing quality in 13 real industrial benchmarks: 5.14%/3.47% in worst/total negative slack

Software Engineer R&D Intern **Hsinchu, Taiwan**
Cadence Inc. - Conformal Team June 2020 - Sept. 2020
- Provided electronic change order (ECO) test cases in a 5-person team for Cadence Conformal's robustness test
- Generated 2000+, 5 kinds of cases in which Conformal acts unusually and proposed 3 improved flows to fix

Publication

- **Yen-Hsiang Huang** and Sung Kyu Lim, "CLK-3D: RC-Calibrated Differentiable Clock-Tree Optimization for 3D ICs," IEEE/ACM International Conference on Computer-Aided Design, 2026.
- **Yen-Hsiang Huang** and Sung Kyu Lim, "Snake-3D: Differentiable Learning for Cross-Tier Logic Path Snaking Optimization in 3D ICs," IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025. [PDF]
- **Yen-Hsiang Huang**, Sai Pentapati, Anthony Agnesina, Moritz Brunion, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024. [PDF]
- Sai Pentapati, Anthony Agnesina, Moritz Brunion, **Yen-Hsiang Huang**, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs", ACM International Symposium on Physical Design (ISPD), 2023. [PDF]

Honors and Awards

Honorable Mention Award in the ACM ISPD contest 2021 March 2021
- 4th place worldwide in the contest.

TSMC scholarship for undergraduate student Sept. 2020 - Jan. 2021
- Awarded to approximately 20 undergraduate students in Taiwan every semester.

The Dean's List Award Spring 2020

- Ranked 1st, with 4.30/4.30 GPA in the semester.